RAYMOND DACOSTA AZADDA

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PROFESSIONAL EXPERIENCE

University of Alaska Fairbanks *Fairbanks, AK*
Data Analyst (Temporary Employee), Center for Alaska Native Health Research *07/2025 - present*

Alaska INBRE (IDeA Network of Biomedical Research Excellence) *Fairbanks, AK*
Core Member, Community-Based Participatory Research (CBPR) Core *02/2026 - present*

University of Alaska Fairbanks *Fairbanks, AK*
Graduate Teaching Assistant *08/2023 - 05/2025*

University of Cape Coast *Cape Coast, Ghana*
Teaching and Research Assistant *11/2022 - 08/2023*

EDUCATION

University of Alaska Fairbanks *Fairbanks, AK*
M.S., Statistics and Data Science *08/2023 - 05/2025*

University of Cape Coast *Cape Coast, Ghana*
B.Sc., Actuarial Science *08/2018 - 11/2022*

SELECTED GRANT-FUNDED RESEARCH PROJECTS

Center for Alaska Native Health Research (CANHR), University of Alaska Fairbanks

Current

Neqpiaput Iinruugut (Our Food is Medicine) *2026 - present*

PI(s) Dr. Andrea Bersamin

Study A tribally-driven Indigenous food sovereignty intervention to prevent coronary heart disease among Yup'ik Alaska Native communities in the Yukon-Kuskokwim Delta region, reconnecting participants with traditional foodways and ancestral knowledge. Funded by NHLBI (R01HL181633-01).

Contributions One of two statisticians on the project, contributing to all quantitative analyses, including multilevel modeling and Bayesian methods.

Alaska Alliance for Community Engagement - Partnership for Action Toward Health (AK ACE - PATH) *2025 - present*

PI(s) Dr. Stacy Rasmus

Study A research initiative implementing localized solutions to promote health and prevent chronic disease in rural Alaska Native communities, prioritizing Indigenous knowledge. Part of the NIH Community Engagement Alliance (CEAL) network, funded by NHLBI (OT2HL158287).

Contributions Member of the project's Biostatistics ("Numbers") Team, contributing to all quantitative analyses, including multilevel modeling, Q-methodology, and Bayesian methods.

Materials AK ACE - PATH project page

Qungasvik (Toolbox)-Healthy Hearts: An Alaska Enrichment Initiative (AK Enrichment) *2025 - present*

PI(s)	Dr. Stacy Rasmus
Study	A culturally grounded, evidence-based intervention promoting traditional Yup'ik diet, physical activity, and social connection for Alaska Native Elders and youth. Part of the NIH Community Engagement Alliance (CEAL) network, funded by NHLBI (OT2HL158287).
Contributions	Member of the project's Biostatistics ("Numbers") Team, contributing to all quantitative analyses, including multilevel modeling, physical activity analysis, and Bayesian methods.
Materials	NIH CEAL Network

Alaska INBRE Community-Based Participatory Research (CBPR) Core *2026 - present*

PI(s)	Dr. Katie Cueva
Study	A core within the Alaska INBRE (IDeA Network of Biomedical Research Excellence) - an NIH program supporting biomedical and behavioral research - that promotes equitable, inclusive, and sustainable community-based participatory research across Alaska. The Core developed a CBPR toolkit (Summer 2025, with a Community Advisory Board) providing guidance, training, and proposal support for researchers and community partners.
Contributions	Conduct semi-structured interviews with CBPR researchers across Alaska, perform thematic analysis of interview transcripts, and help produce one-pagers that synthesize community-engaged research practices into the Core's CBPR Toolkit.
Materials	CBPR Toolkit

Past

Prevention of Incarceration and Recidivism Impacting Alaska Native People *2025 - 2026*

PI(s)	Dr. Katie Cueva
Study	Investigated factors driving the disproportionate incarceration of Alaska Native people (40% of Alaska's prison population despite making up 14% of the state population). Funded by Alaska House Bill 66 (2024) through the Alaska Federation of Natives; data collection included a literature review, over 20 expert interviews, and over 80 surveys.
Contributions	Served as project statistician on the final stakeholder report to the Alaska Legislature, analyzing over 80 survey responses and contributing to thematic analysis of over 20 expert interview transcripts.
Materials	Final Report to the Alaska Legislature

DATA AND SOFTWARE DEVELOPMENT

CANHRActi: A Web-Based Dashboard for the Analysis of ActiGraph Data *2025 - present*

Purpose	A software package and web-based dashboard providing a complete pipeline for accelerometer data analysis in health research. Integrates wear time detection, physical activity classification, sleep scoring, energy expenditure estimation, circadian rhythm analysis, and sedentary fragmentation analysis in a single platform. Includes a novel wear time algorithm for Alaska Native populations and a unique methodology for separating sleep from waking sedentary behavior.
Built with	R, Shiny, C++, CSS; deployed on Posit Connect Cloud
Disclosure	Invention disclosure submitted to UAF Office of Intellectual Property and Commercialization, 04/2026
Link	CANHRActi Dashboard

Intertribal Healthy Data Dashboard *2025 - present*

Purpose	An online platform displaying community-level health information and protective factors. Presents findings from the Native Transformations Project (NTP) - a Tribally initiated, directed, and controlled (NWIC NARCH) research study on strengths, protections, and resilience against substance abuse among the Lummi Nation, Swinomish Indian Tribal Community, and Upper Skagit Indian Tribe (Pacific Northwest). NTP is led by Northwest Indian College (NWIC) and the Center for Alaska Native Health Research (CANHR) at the University of Alaska Fairbanks.
Built with	R, SCSS, JavaScript, Lua, CSS, HTML
Link	Intertribal Healthy Data Dashboard

bayesqm *2026 - present*

Purpose	An R package providing the first fully Bayesian framework for Q-methodology factor analysis. Since Stephenson (1935), Q analysis has relied on frequentist factor extraction with the standard error rule of Brown (1980), which returns participant-to-factor assignments as binary flagged/unflagged decisions and offers no measure of uncertainty. bayesqm replaces that with posterior credible intervals for every loading, probabilistic factor membership $P(\text{dominant}_i = k \mid Y)$ that makes cross-loading visible and quantifiable, and a principled K-selection protocol built on element-wise PSIS-LOO. The model uses a Student-t likelihood for robustness to idiosyncratic Q-sorts and resolves rotational, sign, and label-permutation ambiguity through the MatchAlign post-processing of Poworoznek et al. (2025). Posterior sampling via Stan (cmdstanr or rstan).
Built with	R, Stan
Link	bayesqm

sleepwaker *2025*

Purpose An R package for processing batch sleep export data from ActiGraph ActiLife software. Automates two transformations: (1) reformatting Cole-Kripke and Sadeh sleep-detection output into subject-log diary format for sleep pattern analysis, and (2) inverting detected sleep periods into wake periods to enable sedentary-behavior and physical-activity analysis constrained to waking hours. Solves two recurring problems in 24-hour accelerometry workflows: time-consuming, error-prone manual reformatting, and inflated sedentary-time estimates when sleep periods are misclassified as sedentary behavior. The core innovation is a temporal inversion algorithm that chronologically pairs consecutive sleep periods to compute complement wake periods, producing reproducible separation of sleep from true waking sedentary behavior.

Built with R

Link sleepwaker

canhrcolors *2025*

Purpose A cross-platform color scheme package providing CANHR with a consistent visual identity across data visualizations. Implements twelve named CANHR colors with native integration into R (base graphics, ggplot2) and Stata's scheme system.

Built with R, CSS, Stata

Link canhrcolors

PUBLICATIONS

Peer-Reviewed Journal Articles

Manuscripts Under Review

1. Hueffer, K., Brinkman, T., Robinson, H., Joseph, V., Chaliak, J. A., John, S., Clinton, M., Shaw, J., Peter, T., Bersamin, A., Allen, J., **Azadda, R. D.**, Grogan-Kaylor, A., Lee, K., & Rasmus, S. (2026). Health Priorities for Rural Alaska Native Communities at the Extremes of Weather and Climate Change. *Submitted to BMC Medicine (Springer Nature).*

PRESENTATIONS

Conference Presentations

1. Robinson, H. D., Hueffer, K., Peter, T., Brinkman, T., Clinton, M., Bersamin, A., Charles, W., Joseph, V., John, S., Chaliak, A. J., **Azadda, R. D.**, Allen, J., Lee, K., Grogan-Kaylor, A., Rowles, F., & Rasmus, S. (2026). An adapted version of Q-methodology provides an in-depth understanding of the impact of climate change and health on rural Alaskan communities. *2026 CEAL Annual Meeting.*
2. Rasmus, S., Cueva, K., Robinson, H. D., Grogan-Kaylor, A., O'Brien, D., Bersamin, A., Allen, J., Lee, K., **Azadda, R. D.**, Chaliak, A. J., Johnson, J., Joe, J., Rowles, F., & Shaw, J. (2026). The Qungasvik-Healthy Hearts Intervention: A culturally grounded and evidence-based intervention to support increased physical activity, a traditional Yup'ik diet, and social connection for Elders and youth in Alaska. *2026 CEAL Annual Meeting.*

CONFERENCES, WORKSHOPS, AND TRAINING

EPSCoR Respectful, Cooperative Research Workshop *Fairbanks, AK*
Alaska EPSCoR, University of Alaska Fairbanks *04/2026*

Workshop on Environmental Research Priorities in Interior Alaska *Fairbanks, AK*
University of Alaska Fairbanks *02/2026*

Summer Graduate School: Mathematics of Sea Ice and Polar Ecosystems *Fairbanks, AK*
Simons Laufer Mathematical Sciences Institute (SLMath) *06/2025*

KU Probability and Statistics Conference *Lawrence, KS*
University of Kansas *11/2024*

JSM Diversity Mentoring Program *Portland, OR*
Joint Statistical Meetings, American Statistical Association *08/2024*

Multi-Species Occupancy Model Workshop *Fairbanks, AK*
Alaska Chapter, American Statistical Association *06/2024*

TECHNICAL SKILLS

Programming Languages	R, Stata, Python, C++, JavaScript
Statistical Software & Tools	RStudio, Git/GitHub, LaTeX, JAGS, Stan, KADE, SmartPLS 4, ActiLife
Statistical Methods	Multilevel Modeling; Structural Equation Modeling (SEM); Bayesian Methods; Q Methodology; Machine Learning; Spatial-Temporal Modeling
Qualitative Methods	Thematic Analysis and Q Methodology
Survey Design & Data Capture	REDCap - design and deployment of advanced longitudinal, multi-site survey instruments with branching logic, calculated fields, and custom UI elements (progress bars, visual enhancements) for engaging participant experience; data export and cleaning
Domains	Indigenous and Alaska Native Health; Community-Based Participatory Research (CBPR); Sleep and Physical Activity Research; Behavioral Health; Environmental Health

TEACHING, LEADERSHIP, AND SERVICE

Teaching

University of Alaska Fairbanks *Fairbanks, AK*
STA 401: Regression and Analysis of Variance (TA + Lab Instructor) *2024 - 2025*
Math 252: Calculus 2 (Teaching Assistant) *2023*
STA 200X: Elementary Statistics (Teaching Assistant) *2023*

University of Cape Coast *Cape Coast, Ghana*
STA 406: Time Series Analysis (Teaching Assistant) *2022*
STA 301: Probability Distributions (Teaching Assistant) *2022*
STA 101: Introduction to Probability (Teaching Assistant) *2022*

Leadership

University of Cape Coast *Cape Coast, Ghana*
Educational Head, Actuarial Club *2022*
Launched and coordinated the Empowered Data Analyst Competition - a semester-end data analysis and visualization showcase that promoted statistical applications to local health, agriculture, and family welfare research.

Class Representative, Actuarial Science Students *2018 - 2022*
Facilitated communication between students and faculty and disseminated departmental updates and decisions.

Service

University of Cape Coast *Cape Coast, Ghana*

Co-Author, Departmental Solution Manual *2021 - 2023*

Co-authored solution manuals for past examination questions in the department's actuarial science and statistics courses: ACT 212 (Mathematics of Investment and Credit), ACT 214 (Financial Analysis and Economics), ACT 301 (Life Contingencies I), ACT 304 (Life Contingencies II), and STA 306 (Multivariate Distributions).

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA) *2024 - present*

Institute of Mathematical Statistics (IMS) *2024 - present*

Society of Actuaries (SOA) *2022 - present*